



CS301-FALL
ASSIGNMENT-1
JUNAID MALIK



Junaidfzal08@gmail.com
Bc190202640@vu.edu.pk

FOR MORE VISIT
VULMSHELP.COM

JUNAID MALIK
0304-1659294



CS619 & CS519 PROJECTS

پروجیکٹ اور اسٹریٹج
ساتھ ساتھ

Daily Classes & Viva Preparation

Our Expertise

- ✓ Python (Django & Flask)
- ✓ Php (Laraval)
- ✓ MySql

Projects Phase

- ✓ SRS + DD Prototype
- ✓ Final Deliverable



LMS Handling

+92 304 1659294

ajtechinstitute24@gmail.com

AL-JUNAID TECH INSTITUTE

JOIN OUR GROUP

AL-JUNAID TECH:

<https://chat.whatsapp.com/B5ZQwE3mkkC1KHYUeA0uji>

CS619-CS519:

<https://chat.whatsapp.com/DqFmAIrPZdJHhpxAsvEbVf>

QUESTION NO 1:

SOLUTION:

```
#include <iostream>
```

```
using namespace std;
```

```
class Product {
```

```
private:
```

```
    int data;
```

```
    Product* next;
```

```
    int productId;
```

```
public:
```

```
    // Get and Set functions for `data`
```

```
    int get() { return data; }
```

```
    void set(int data) { this->data = data; }
```

```
    // Get and Set functions for `next`
```

```
    Product* getNext() { return next; }
```

```
    void setNext(Product* nextNode) { this->next = nextNode; }
```

```
    // Get and Set functions for `productId`
```

```
    int getProductId() { return productId; }
```

AL-JUNAID TECH INSTITUTE

```
void setProductId(int pid) { this->productId = pid; }
```

```
};
```

```
class Stack {
```

```
private:
```

```
    Product* head;
```

```
public:
```

```
    Stack() : head(nullptr) {}
```

```
// Push function to insert an element
```

```
void push(int x) {
```

```
    Product* newProduct = new Product();
```

```
    newProduct->set(x);
```

```
    newProduct->setNext(head);
```

```
    head = newProduct;
```

```
    cout << "Pushed digit " << x << " onto the stack." << endl;
```

```
}
```

```
// Top function to get the top element of the stack
```

```
int top() {
```

```
    if (head != nullptr) {
```

```
        return head->get();
```

```
    }
```

```
    cout << "Stack is empty!" << endl;
```

```
    return -1; // Indicating empty stack
```

```
}
```

```
// Pop function to remove the top element from the stack
```

AL-JUNAID TECH INSTITUTE

```
int pop() {  
    if (head == nullptr) {  
        cout << "Stack is empty!" << endl;  
        return -1; // Indicating empty stack  
    }  
    int poppedValue = head->get();  
    Product* temp = head;  
    head = head->getNext();  
    delete temp;  
    return poppedValue;  
}  
};  
  
int main() {  
    Product product;  
    Stack stack;  
  
    // Set the product ID by combining "24" with last two digits of student ID  
    int lastTwoDigits = 40; // Replace with actual last two digits of your student ID  
    int productId = 2400 + lastTwoDigits;  
    product.setProductId(productId);  
  
    cout << "-----" << endl;  
    cout << "JUNAID MALIK " << endl;  
    cout << "03041659294 " << endl;  
    cout << "-----" << endl;  
    // Display Product ID  
    cout << "The Product id is: " << product.getProductId() << endl;  
    cout << "-----" << endl;
```

AL-JUNAID TECH INSTITUTE

```
// Break down the Product ID into individual digits in reverse order and push onto the stack
```

```
int tempId = product.getId();
```

```
int digits[4]; // assuming a 4-digit product ID
```

```
int i = 0;
```

```
while (tempId > 0) {
```

```
    digits[i++] = tempId % 10;
```

```
    tempId /= 10;
```

```
}
```

```
// Push digits onto the stack in reverse order (to match your expected output)
```

```
for (int j = i - 1; j >= 0; j--) {
```

```
    stack.push(digits[j]);
```

```
}
```

```
cout << "-----" << endl;
```

```
// Display the top element of the stack
```

```
cout << "The Top element of the stack is: " << stack.top() << endl;
```

```
cout << "-----" << endl;
```

```
// Pop two elements and display them
```

```
cout << "The First element popped from the stack is: " << stack.pop() << endl;
```

```
cout << "The Second element popped from the stack is: " << stack.pop() << endl;
```

```
// Display the new top element of the stack
```

```
cout << "The Top element of the stack is: " << stack.top() << endl;
```

```
return 0;
```

AL-JUNAID TECH INSTITUTE

}

OUT PUT:

JUNAID MALIK
03041659294

The Product id is: 2440

Pushed digit 2 onto the stack.
Pushed digit 4 onto the stack.
Pushed digit 4 onto the stack.
Pushed digit 0 onto the stack.

The Top element of the stack is: 0

The First element popped from the stack is: 0
The Second element popped from the stack is: 4
The Top element of the stack is: 4

JOIN OUR GROUP

AL-JUNAID TECH:

<https://chat.whatsapp.com/B5ZQwE3mkkC1KHYUeA0uji>

CS619-CS519:

<https://chat.whatsapp.com/DqFmAIRPZdJHhpxAsvEbVf>